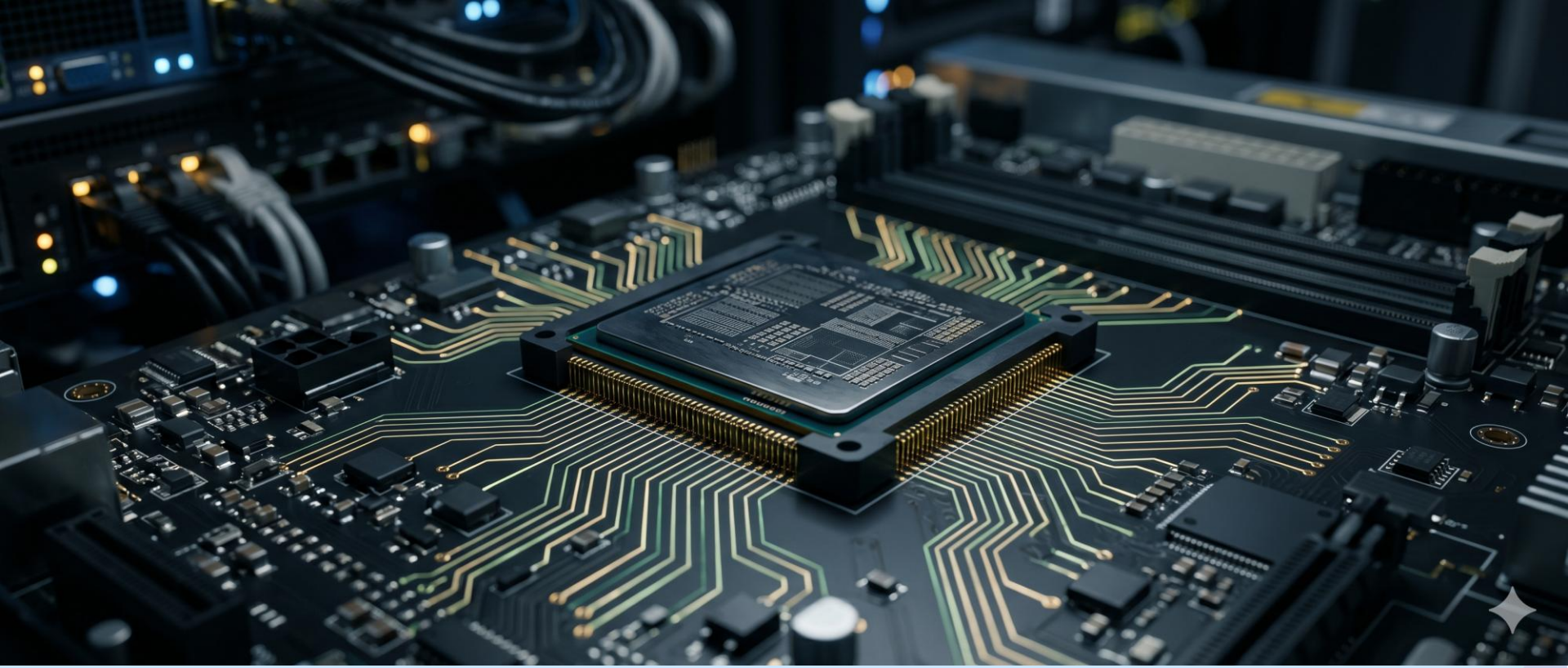


Beyond BMC: Architecting the Future of Servers

Chris Lin

May 2026

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Safe Harbor Notice

ASPEED's statements of its current expectations are forward-looking statements subject to significant risks and uncertainties and actual results may differ materially from those contained in the forward-looking statements. Except as required by law, we undertake no obligation to update any forward-looking statement, whether as a result of new information, future events, or otherwise.

Please note that while we have made every effort to list all significant risks, there may still be other risk factors that have not been identified or currently appear to be insignificant.

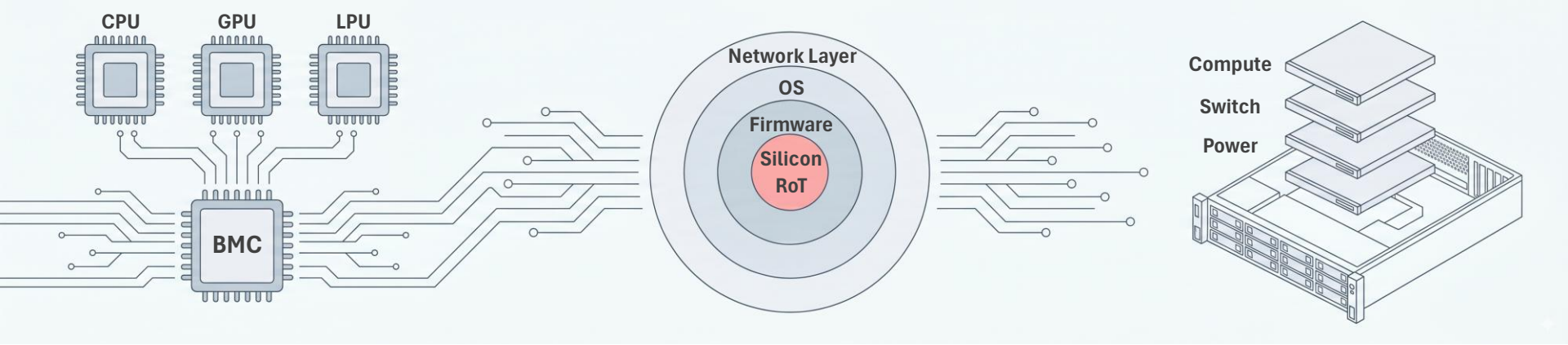
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請注意，儘管我們已盡力羅列各項重要風險，但仍可能存在尚未識別或當前看似不顯著的其他風險因素。

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伺服器設計主要驅勢



Heterogeneous Computing

- ✓ Increasing complexity
- ✓ XPU collaboration

Zero Trust & Root of Trust

- ✓ Platform RoT
- ✓ Silicon RoT
- ✓ PUF & PQC

Modular Design (Disaggregated)

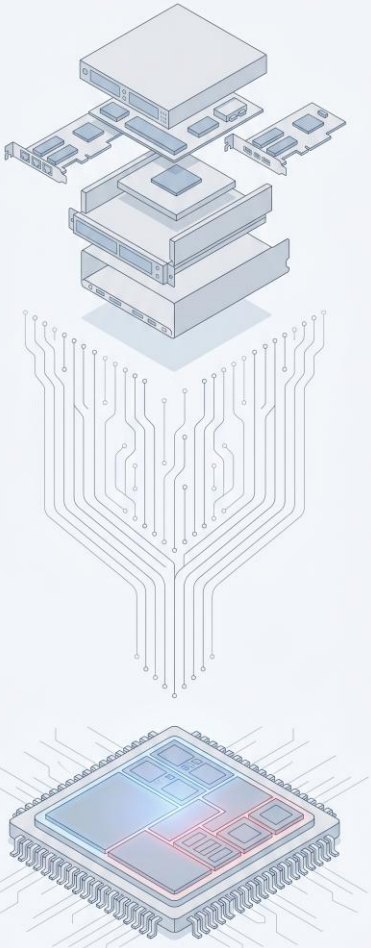
- ✓ Maintenance and upgrade
- ✓ Time-to-market
- ✓ Space efficiency
- ✓ Flexible manufacturing

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信驊科技經營策略

- Maximize *dollar content per server board with product suite;*
- Accelerating *product cadence.*

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信驊產品的策略優勢

BMC: Setting the Standard

✓ Co-designed with hyperscalers and chip makers

SMC*: Multi-Card Architectures

✓ Empowering extreme scalability with efficient 2-level management

PRoT: Security for Platform

✓ Protection, detection, recovery

I/O Expander: Seamless Integration

✓ Bridging decoupled modules

*We further enhance security by integrating **Caliptra** into new generation of all chips.*

Note: Satellite Management Controller (SMC) is also known as Auxiliary Management Controller (AMC) or Bridge IC

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極致靈活性：將 FPGA 嵌入系統單晶片

Partnership: Collaborating with FPGA leader **Lattice**.

Synergy: Embedding FPGA logic directly into AST1840 SoC

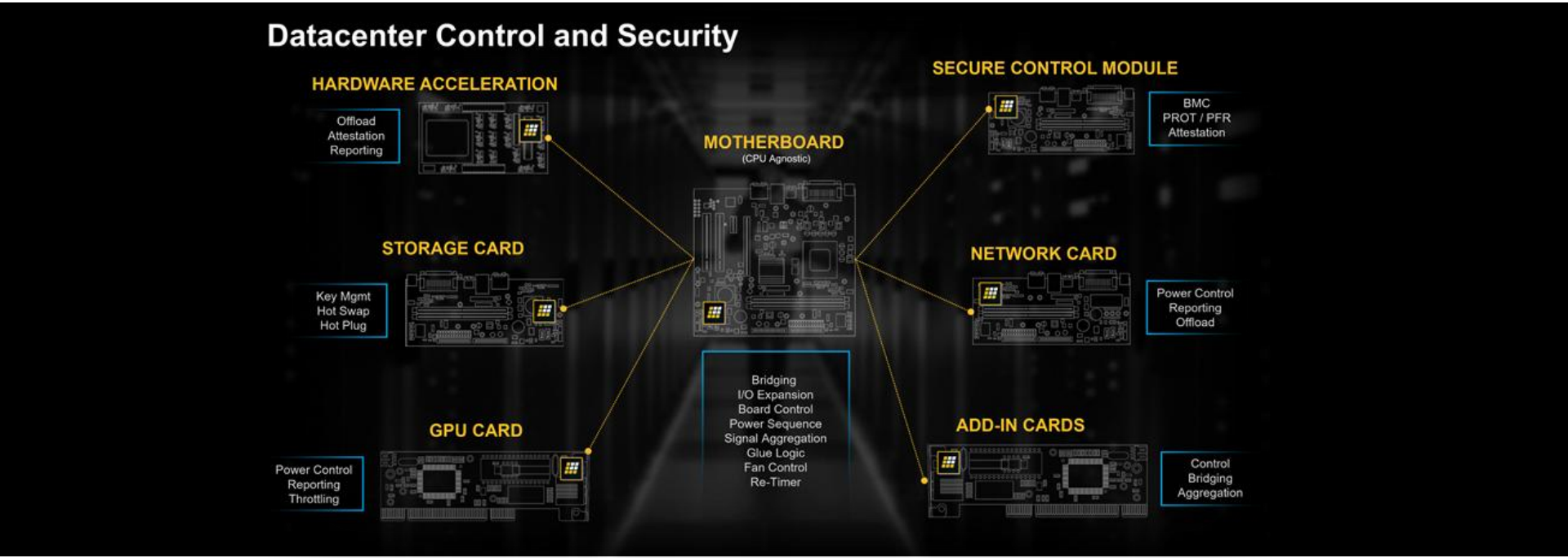
Flexibility: Delivering instant-on-programmable control for evolving server needs



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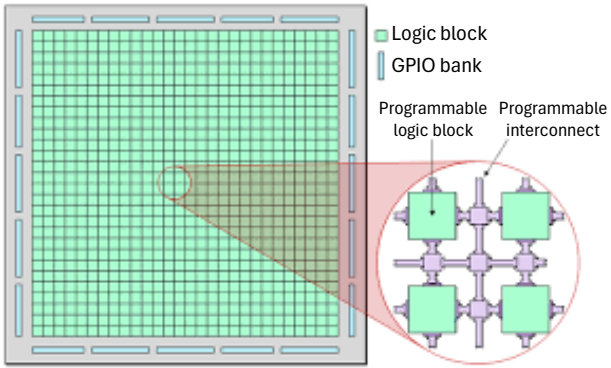
Lattice FPGA 的各種應用

Lattice FPGA has multiple applications in datacenter and is working closely with ASPEED BMC.

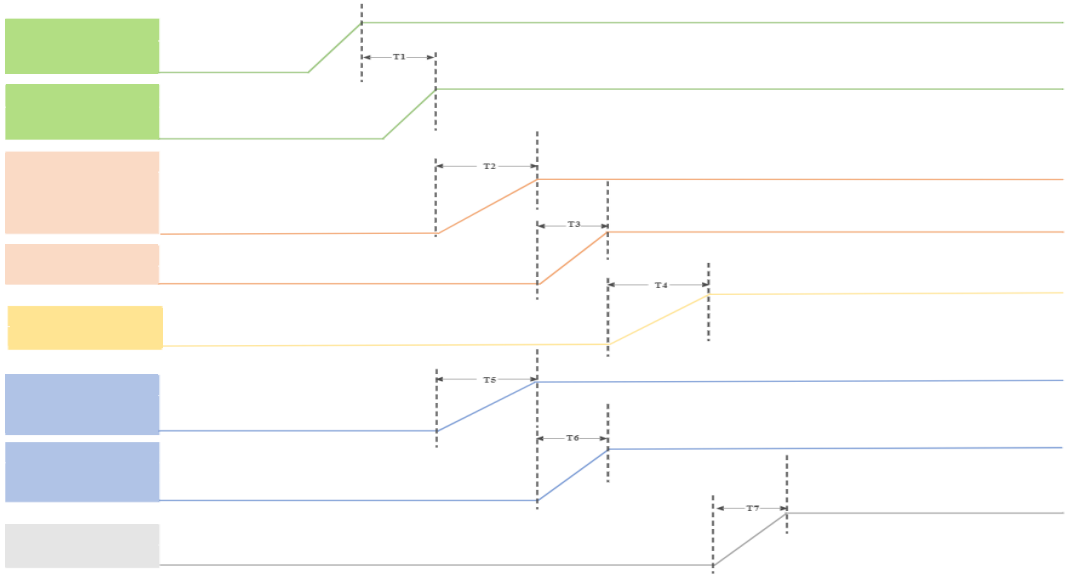


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用FPGA 做電源時序管理



Top view of FPGA



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Lattice 的優勢

Instant-On Control

✓ Zero-delay execution for precise power sequencing and system resets

Agile Customization

✓ Programmable logic to rapidly adapt to diverse server SKUs

Parallel Real-time Processing

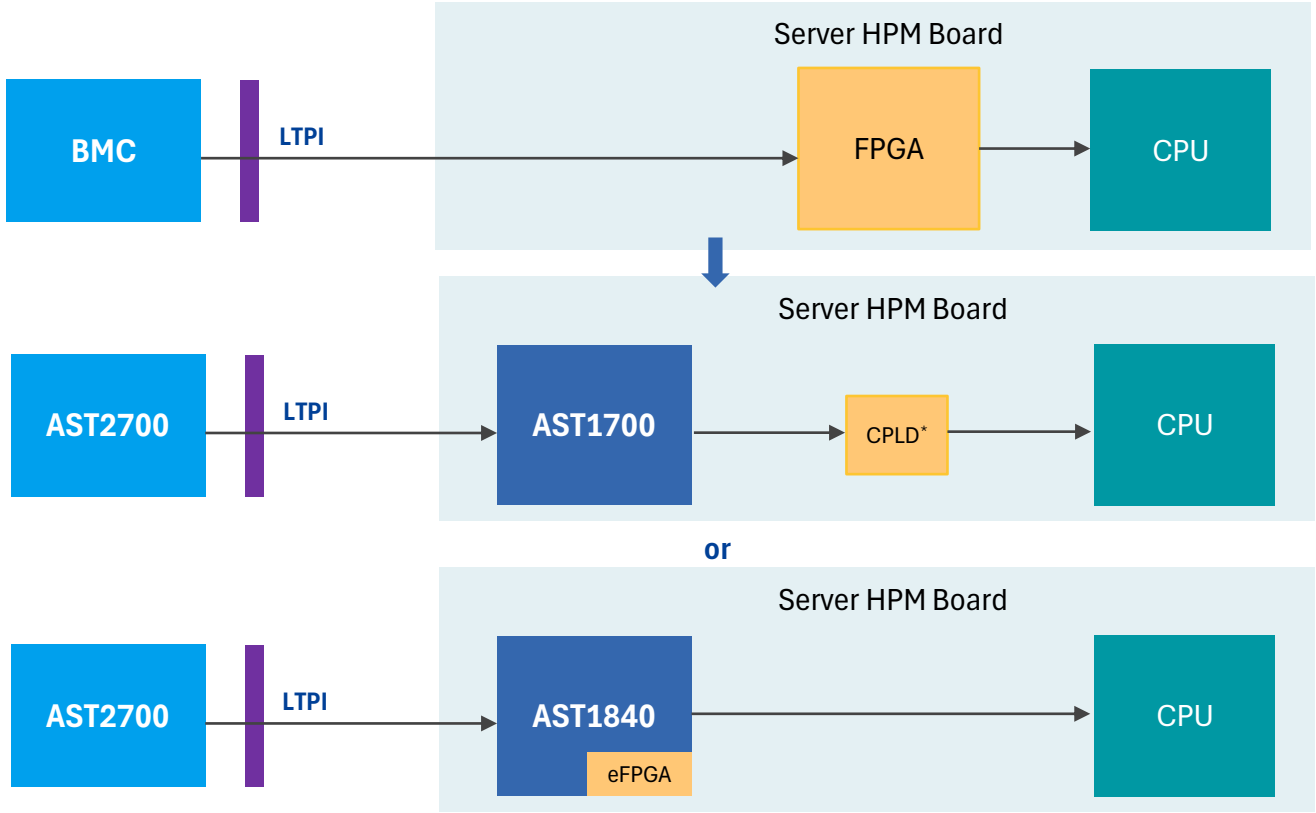
✓ Manage power, telemetry, and I/O tasks wit no firmware bottlenecks

I/O Consolidation & Bridging

✓ Unifying diverse interfaces (I2C, SPI) to simplify complex board designs

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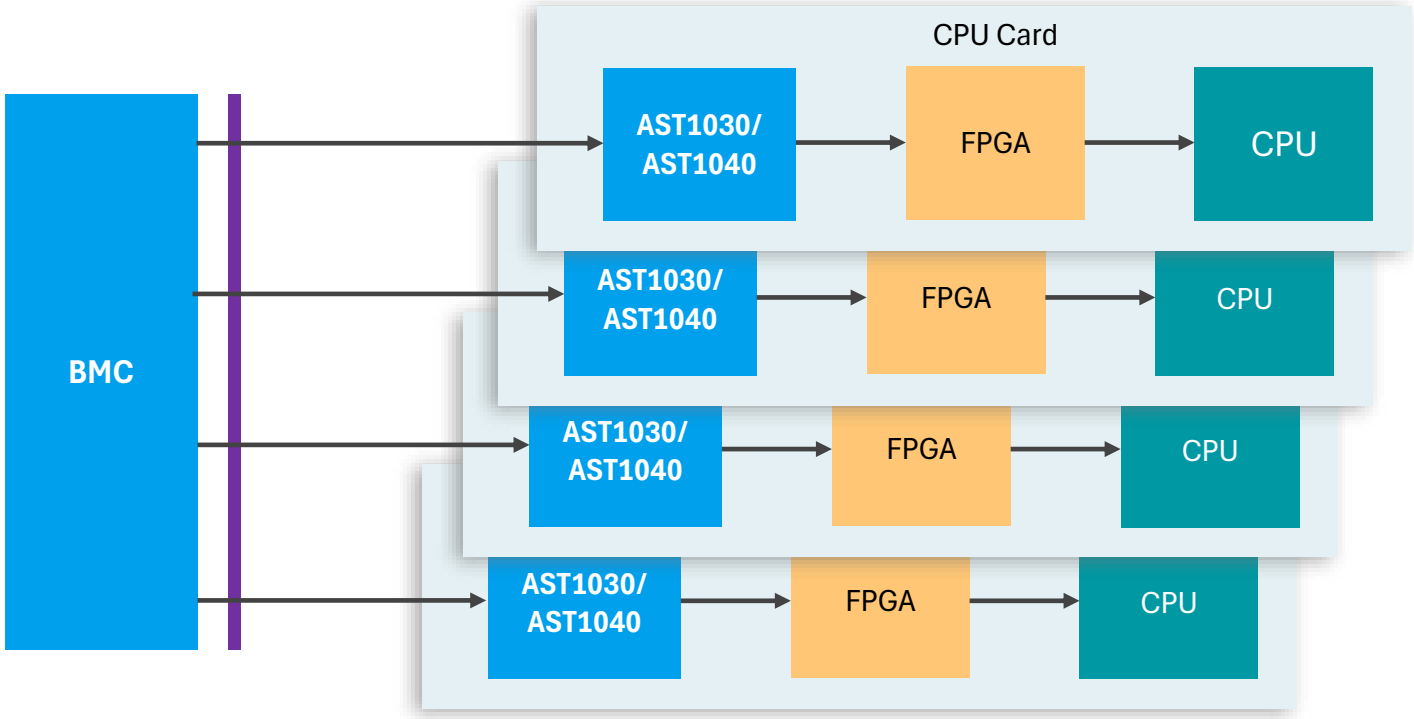
持續演進的伺服器架構 (單節點)



Note: CPLD = small FPGA

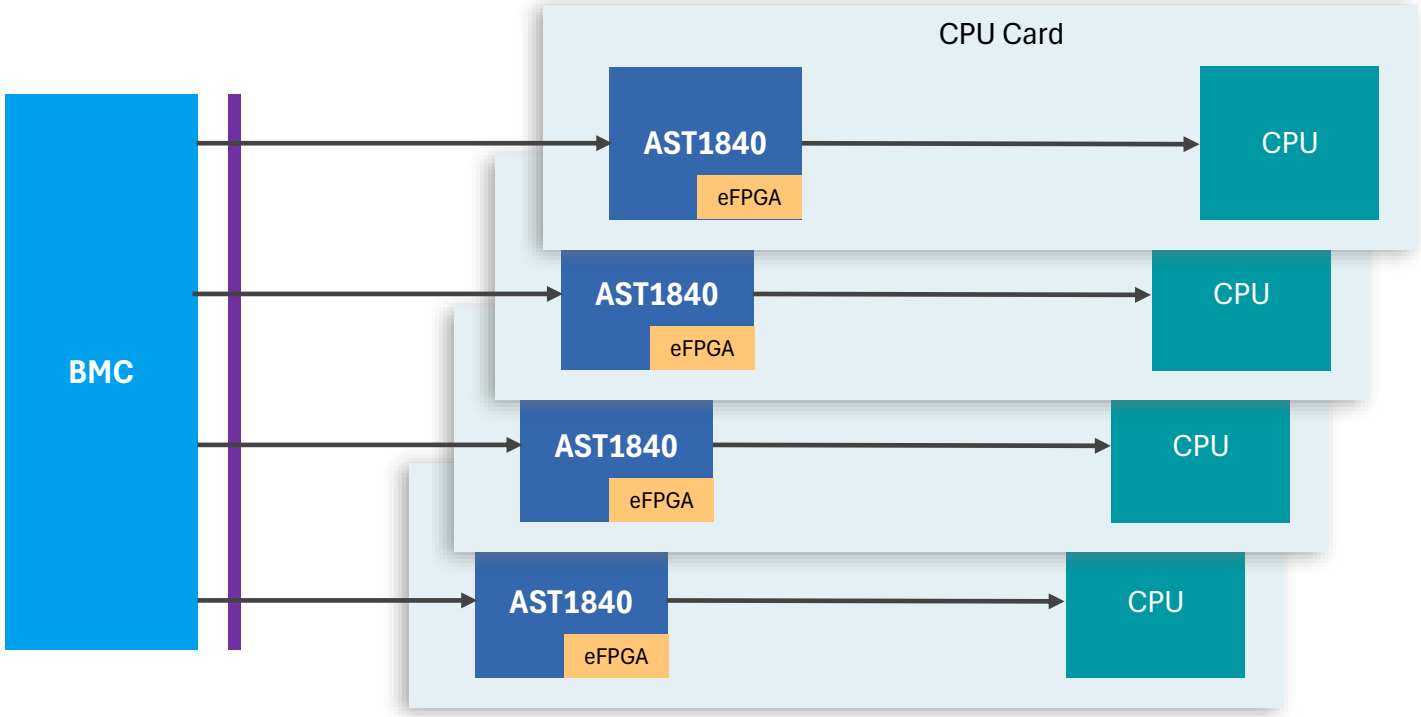
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多節點架構 (含 FPGA)



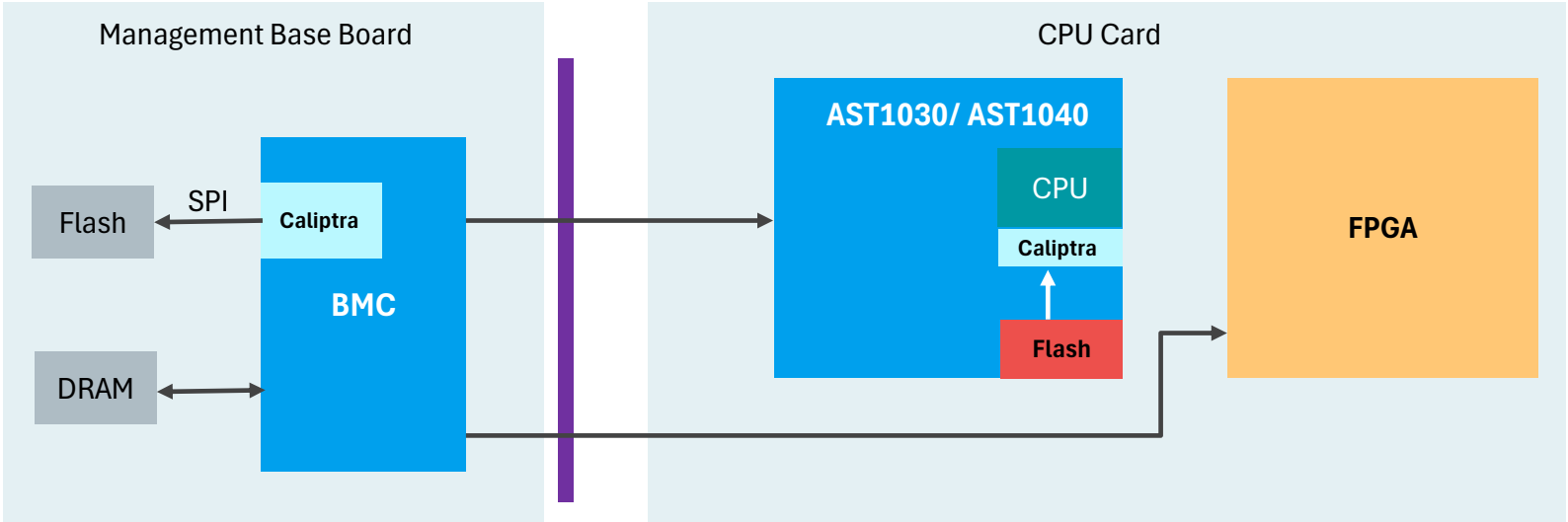
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多節點架構簡化設計 (無 FPGA)



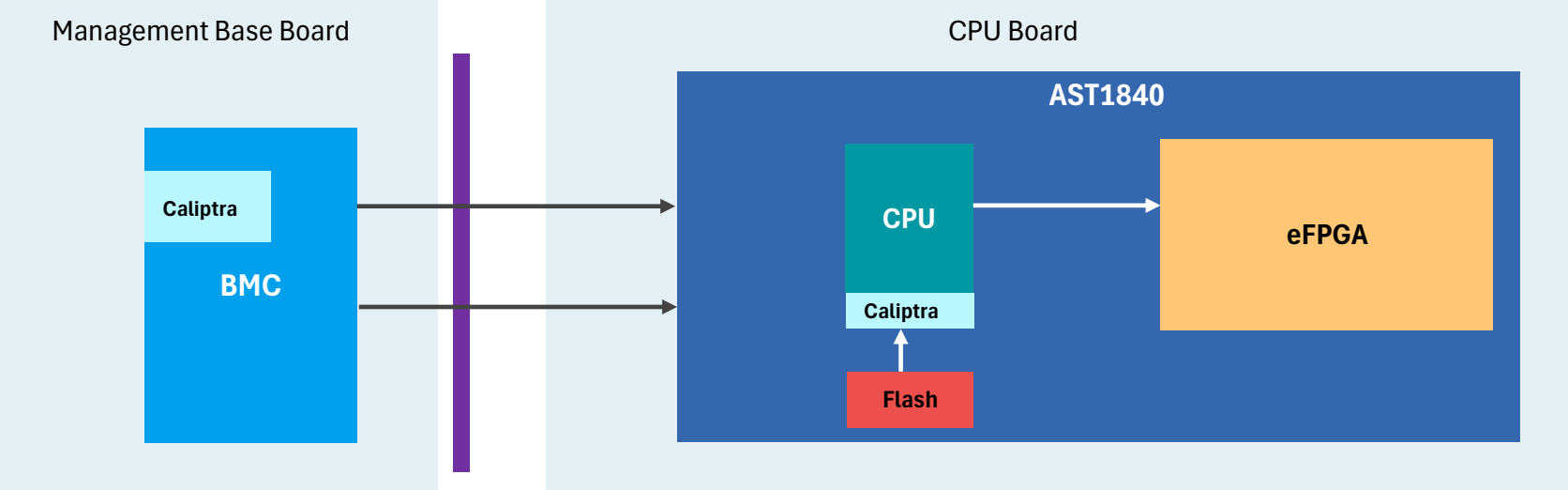
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搭載 FPGA 的伺服器 CPU 板



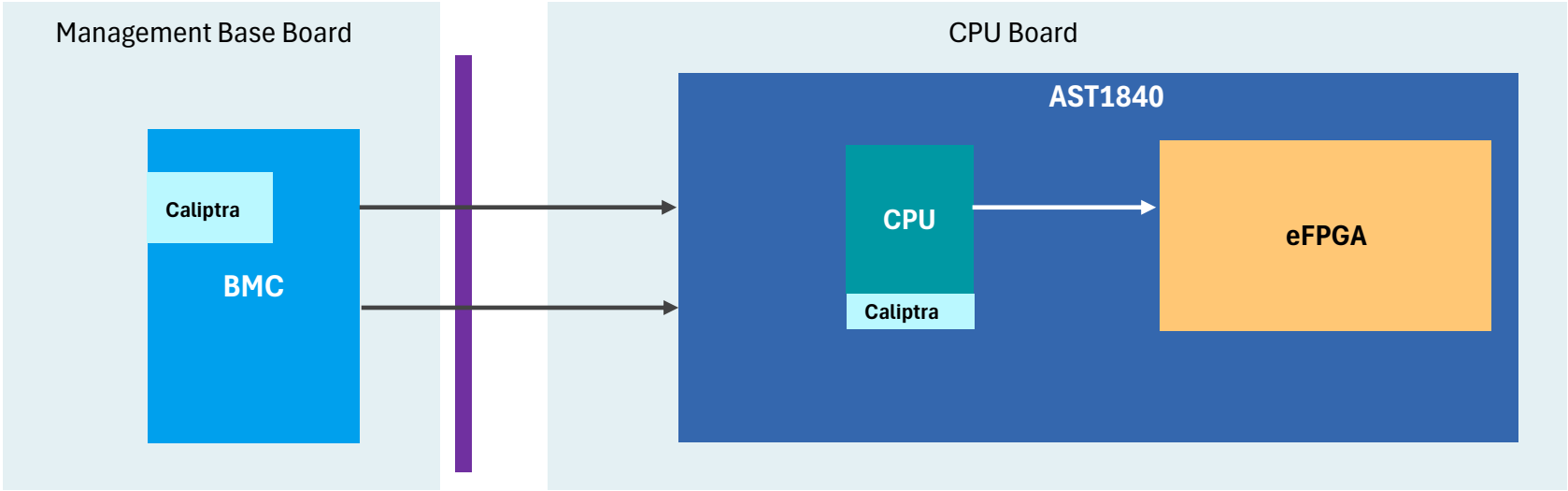
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採用 Flash 的非串流開機



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毋須 Flash 的串流開機



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AST1840 的核心價值

- ✓ **Enhanced Security:** *Sharing Caliptra with eFPGA*
- ✓ **Simplified Design:** *No external FPGA required*
- ✓ **Versatile:** *Suitable for various server designs*
- ✓ **Streaming Boot:** *Easy Update*

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時程與潛在市場規模

Schedule: *Production-ready in 2027 Q3*

Potential TAM: *Multi million units*

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ASPEED Leading the Architecture Transformation

Beyond BMC

Driving the shift from single-node to multi-node heterogeneous computing.

The Best of Two Worlds

Maximizing product value with ASIC efficiency and eFPGA adaptability.

Streamlined and Agile

Simplifying design while ensuring hardware readiness for future upgrades.

Massive Potential

Expanding our TAM alongside next-gen server architectures.

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Q&A

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